

STEPHANE BANCELIN

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Laboratoire Photonique Numérique et Nanosciences (LP2N)

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CNRS RESEARCHER IN BIOPHOTONICS

SCIENTIFIC AREAS

- Light/Matter interaction, biophotonics, nonlinear optics, neuroscience
- Optical microscopy: multiphoton (SHG, CARS, THG, SFG/DFG), super-resolution (STED), polarimetry, ellipsometry, interferometry, adaptive optics
- Computational analysis: 3D reconstruction, segmentation, morphological mathematics, numerical modelling
- Collagen chemistry: protein aggregation (fibrillogenesis), extraction/purification, hydroxyproline dosing
- Tissue manipulation & imaging: skin, brain, bone, cartilage, cornea, embryos, tendon, biomimetic materials
- Tissue physiopathology: skin biomechanics, prolapses, cancer, embryogenesis, EDS, osteoarthritis, Alzheimer
- Animal experiments: in vivo imaging, brain surgery, stereotaxic injection, animal care

RESEARCH EXPERIENCES

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| Since 2023 | CNRS Researcher
Laboratoire de Photonique Numérique et Nanosciences (LP2N), CNRS - Institut d'Optique - Univ. de Bordeaux
<i>Non-linear microscopy to study connective tissue biomechanics.</i> |
| 2020-2022 | CNRS Researcher
Institut Interdisciplinaire de Neurosciences (IINS), Université de Bordeaux - CNRS
<i>Adaptive optics for in vivo STED microscopy.</i> |
| 2016 - 2019 | Post-doctoral researcher with Prof. V. Nägerl
Institut Interdisciplinaire de Neurosciences (IINS), Université de Bordeaux - CNRS
<i>In vivo super-resolution STED imaging of dendritic spines in the hippocampus.</i> |
| 2014 - 2016 | Post-doctoral researcher with Prof. F. Légaré
Institut National de la Recherche Scientifique, Centre Energie Matériaux et Télécommunication, Université du Québec
<i>Interferometric Second Harmonic Generation microscopy for non-invasive tissue imaging.</i> |
| 2010 - 2013 | PhD in Biophotonics under the supervision of Dr Marie-Claire Schyenne-Klein
Laboratoire d'Optique et Biosciences (LOB), École Polytechnique - CNRS – Inserm
<i>Quantitative imaging of collagen using Second Harmonic Generation microscopy.</i> |

2010	Master internship with Dr K. Plamann Laboratory for Applied Optics (LOA), ENSTA ParisTech - CNRS <i>Study of corneal transparency</i>
2009	Master internship with Dr David Neely Rutherford Appleton Laboratory (RAL), University of Oxford <i>Protection of the optics in the HIPER inertial fusion reactor</i>
2008	Bachelor internship with Anraud Landragin Laboratoire de référence Temps/Espace (SYRTE), Observatoire de Paris <i>Implementation of an extended cavity laser : application to cold atoms manipulations</i>

TRAINING

2013	PhD in Biophotonics Ecole Polytechnique
2013	University degree of advanced biomedical studies IFSBM - Paris Sud University
2010	MSc in Optics with a specialty in Laser/Matter interactions <i>ENS Cachan - Ecole Polytechniques - Paris VI University (UPMC)</i>
2008	BSc in experimental physics <i>ENS Cachan - Paris VI University (UPMC)</i>

TEACHING

Since 2020	Coordinator of the IINS-BIC Labcourse for the M2 EUR-Light program at the University of Bordeaux (2 weeks/year)
2021-2024	Master's courses : EUR Light et EUR Neurophotonic (6 h/year). <i>Super-resolution STED microscopy in neurosciences et Microscopy beyond fluorescence.</i>
Since 2023	Bibliographic projects at the Institut d'Optique (1 week/year)
2018 - 2023	Instructor for the National Training Action - Bordeaux Imaging Center (~4 h/year) <i>Microscopie de super-résolution.</i>
2017 - 2022	Instructor at the CAJAL School , Bordeaux School of Neuroscience (2 weeks/year). <i>Connectomics : from micro to meso and macro-scale, Linking neuronal circuit and behavior, Advanced imaging techniques for cellular and systems neuroscience.</i>
2010 - 2013	Teaching assistant in physics, Paris-Sud Faculty of Medicine (64h/year)
2008 - 2010	Oral examiner for the preparatory class for Engineer school, PSI* track - Lycée Jean-Baptiste Say (Paris) (3h/week)

STUDENT'S SUPERVISION

○ PhD students	
Since 2026	Florestan Luga , PhD student at LP2N, Co-direction with F. Légaré at the University of Quebec <i>Correlative SHG - Brillouin microscopy in articular cartilage.</i>
Since 2025	Théo Martinaud , PhD student at LP2N <i>Nonlinear optical microscopy to visualize the architecture of connective tissues.</i>
Since 2023	Nasim Akhtar , PhD student at LP2N, Co-direction with L. Cognet <i>Multimodal 3-photon fluorescence microscopy applied to neurophotonics.</i>
Since 2023	Rohit Mangalwedhekar , PhD student at LP2N, Co-direction with L. Cognet <i>Super-resolution multimodal dynamic microscopy in biophysical models and biological tissues.</i>
2020 - 2023	Johannes Roos , PhD student at IINS - Co-supervision with V. Nägerl <i>Arkitekt: An open-source framework for modern bioimaging workflows.</i>

⇒ Currently post-doctoral researcher at Johannes Kepler University of Linz

○ **Undergrad student**

2025	Théo Martinaud , Master's student at LP2N <i>Implementation of a VIPA spectrometer and integration into a scanning microscopy.</i>
2025	Matys Lambert , 3 rd - year in the IUT Physical measurements program <i>Building of a symmetric tensile extensometer to study connective tissues.</i>
2023	Gioele Sassi , M1 student at LP2N - currently PhD student at the University of Trento <i>Design and implementation of a microscope for Multiphoton imaging of Single-Walled Carbon Nanotubes in IR range.</i>
2022	Mohamed Belkadi , M2 student at IINS - currently engineer at Belimo <i>PSF optimization using SLM-based adaptive optics in brain tissue.</i>
2021	Philippe Nguyen , M2 student at IINS - currently PhD at CRPP <i>Development of quantitative approaches for inverse-contrast STED microscopy.</i>
2019	Sabin Sapkota , Bachelor student at IINS <i>Measuring laser pulse duration in a STED light using Michelson interferometer.</i>
2018	Carlotta Pessina , M2 student at IINS - currently lead consultant at BIP <i>2P-STED microscopy and aberration correction through adaptive optics for neuroscience applications.</i>
2017	Rached Baida , M1 student at IINS - currently engineer at Seemens Healthinners <i>Confocal microscopy with a Tip-tilt scanner.</i>
2015	Etienne Auger , Bachelor student at INRS-EMT - currently engineer at IR&T <i>Interferometric SHG imaging of equine cartilage.</i>
2014	Katherine Légaré , Bachelor student at INRS-EMT - currently researchers at TeraXion <i>Image analysis in interferometric SHG microscopy.</i>

COLLECTIVE RESPONSIBILITIES & REVIEWING ACTIVITIES

2026	Expert evaluator for Horizon Europe – Doctoral Network
Since 2025	LP2N Lab council member
2024 - 2025	Organizer of the LP2N microscopy Journal club
Since 2024	Member of the AI comitte, deployment of a local AI on the lab's computing server
2024	Project evaluator - Envergure-Amorçage calls for proposals: Région Bourgogne Franche Comté
2024	Project evaluator - ANR PRC
2024	Organizer of the LP2N laboratory retreat (~80 participants)
2023 - 2024	Coordinator of LP2N student seminars
2023	Reviewer for the Pierre Favard prize – French society of microscopy
2023	Reviewer for the « Espoir en tête » exceptional project call - FRC
2022	Reviewer for the AAP Research Equipement program - A. Midex
Since 2021	Expert evaluator for Horizon Europe - Marie Slodowska-Curie Action (MSCA)
2019 - 2022	Module coordinator at the CHRONIC animal facility at Neurocampus and member of the SBEA animal welfare committee for the PIV-EXPE and CHRONIC animal facilities
Since 2015	Reviewer for international peer-review journals <i>Nature Communication, Optica, Scientific Reports, Biomedical Optics Express, Optics Letters, Optics Express, International Journal of Biological Molecules, Revue Française d'Histotechnologie, NanoLetters, and Scientific Data.</i>
2011 - 2013	Elected member of the LOB laboratory council
2010	Organizer of the 22 nd IFSBM Symposium <i>Nanotechnologies and Biomedical applications</i> (150 participants)

FUNDINGS & DISTINCTION

2026	SMR Departement Call for proposals - 1 year - 7.5k€ Role: PI <i>BriTis: Brillouin microscopy to characterize soft connective tissues biomechanics.</i>
2024	New Aquitaine Research Call for proposal - 3 years - 192 k€ Role : PI Partner : Poietis <i>MOMI : Multimodal optical microscopy to probe the architecture of bioprinted tissues.</i>
2023	ANR JCJC - 5 ans - 349 k€ Role : PI <i>HaBlm: Second Harmonic and Brillouin imaging to probe multiscale vocal cords biomechanics.</i>
2023	Emergence INP - 1 year - 60 k€ Role: PI <i>Setup of a 1700 nm multiphoton microscope for imaging biological tissues.</i>
2023	SMR Departement Call for proposals - 1 year - 10k€ Role: PI <i>BioMulR: Assembly of a microscope-mounted extensometer for studying the biomechanics of connective tissues.</i>
2023	GPR Light - PhD grant for Rohit Mangalwedhekar - 3 years - 115 k€ Role: co-PI with Laurent Cognet (50%) <i>RoMan: Multimodal dynamic super-resolution microscopy in biophysical models and biological tissues.</i>
2020	ERC Synergy - 6 years - 10 M€ PI: Laurent GROC Partners: Lauren COGNET, Erwan BEZARD, Valentin NÄGERL Role: Task leader in Valentin Nägerl during 2021-2022 and in Laurent Cognet since 2023. <i>ENSEMBLE: Structure and function of the brain extracellular space</i>
2018	Marie Curie Individual Fellowship - 2 years - 185 k€ <i>IVSTED: in vivo super-resolution imaging of synapses in the hippocampus.</i>
2017	AXA Research Fund postdoctoral grant - 2 years - 130 k€ <i>Super-resolution STED microscopy in the hippocampus in vivo to reveal changes in synapses underlying memory impairment in Alzheimer's disease.</i>
2014	RTRA Physics Triangle best PhD award
2010	MERNT doctoral grant - 3 years - 90 k€ <i>Quantitative imaging of collagen using Second Harmonic Generation microscopy.</i>